

NORTHGATE ACADEMY
Year 10 Combined Science - Biology Paper 1
Cell Biology, Transport and Respiration

Candidate: A. Whitfield
Date: 12 June 2026
Total marks available: 20

Candidate number: 4172
Time allowed: 40 minutes

Answer ALL questions in the space provided.

- 1 Name the organelle that carries out aerobic respiration, and state the molecule it produces. [2 marks]

Answer: The mitochondrion. It makes ATP.

- 2 (a) Write the word equation for aerobic respiration. [1 mark]

Answer: glucose + oxygen -> carbon dioxide + water

- 2 (b) Explain why muscle cells contain a large number of mitochondria. [2 marks]

Answer: Because muscle cells need a lot of energy, especially when you are exercising.

- 3 A cell is 50 micrometres long. In a drawing of the cell, the same cell measures 100 mm. Calculate the magnification of the drawing. Show your working. [3 marks]

Answer: 50 micrometres = 0.05 mm

$$\begin{aligned}\text{magnification} &= \text{size of image} / \text{size of real object} \\ &= 100 / 0.05 \\ &= 200\end{aligned}$$

Magnification = x200

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- 4 Describe two ways a root hair cell is adapted for taking up mineral ions from the soil. [2 marks]

Answer: It has a long thin extension that sticks out into the soil, which increases the surface area. It also has lots of mitochondria to provide the energy for taking minerals in by active uptake.

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- 5 State one function of the cell membrane. [1 mark]

Answer:

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- 6 Describe how the structure of an alveolus makes it efficient for gas exchange. [4 marks]

Answer: The alveoli are extremely well designed for the job they do and they are one of the most impressive structures in the whole of the human body. There are millions of them in each lung, which gives the lungs a huge surface area for gases to pass across. Each alveolus is also surrounded by a thick layer of muscle which squeezes the oxygen into the blood so that it gets there quickly. Because the alveoli are kept moist, gases can dissolve first and then diffuse.

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- 7 Define osmosis. [3 marks]

Answer: Osmosis is when water moves from a dilute solution to a more concentrated solution.

8 Name the product of anaerobic respiration in human muscle cells and
state one effect it has on the muscle. [2 marks]

Answer: see extra answer space below.

EXTRA ANSWER SPACE

Write the question number next to any answer continued here.

Q8: Lactic acid is made. It builds up in the muscle and causes
muscle fatigue, so the muscle stops contracting as well as it
did before.

END OF PAPER